Kajol Dave

(413) 315 1114 | kajoldave031@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Data Scientist with 5 years of experience building propensity models, segmentation frameworks, and causal measurement systems across financial services, streaming, and market intelligence for Fortune 500 clients

Education

Master of Science in Business Analytics, University of Massachusetts Amherst — Aug 2024 - May 2026
Data Analytics Track — (GPA: 4.0/4.0)
Master of Business Administration, Maharashtra Institute of Technology — Jul 2017 - May 2019
International Business Track
Bachelor of Computer Application, Savitribai Phule Pune University — Jul 2014 - May 2017

Skills

Tools & Languages: Python, R, PySpark, SQL, Azure Databricks, Azure SQL, Power BI, Streamlit, Git, MLflow, Docker
Methods: Machine Learning, ETL, Data Pipeline, XGBoost, Random Forest, LightGBM, Logistic Regression, k-Means, Causal Inference, Latent Class Analysis, RFM, BG/NBD, NLP (TF-IDF, LDA), SHAP, ARIMA, Bayesian Inference, A/B Testing, Sequential Testing (mSPRT), Thompson Sampling, Power Analysis, Exposed-Unexposed Studies, Stakeholder Management, Cross-functional

Experience

Data Scientist - Bajaj Finance Ltd — Mar 2023 - Jun 2024
Azure Databricks (PySpark, MLflow), Azure SQL, Power BI, CDP, Python, SQL, Excel

- Lifted lead conversion by 8% by building product-level cross-sell ML models (XGBoost, Random Forest) scoring 800K+ customers across personal loan, credit card, and gold loan products, contributing 15% of total campaign revenue
- Captured 78% of business volume in the top 3 deciles by developing an RFM-based segmentation and behavioral clustering pipeline ranking customers by conversion likelihood
- Engineered a recommendation engine integrating transaction and Account Aggregator data via CDP, enabling hyper-personalized targeting across SMS, push notification, app banner, and telesales channels
- Designed champion-challenger framework monitoring AUC decay and PSI drift across propensity model variants, triggering automated retraining on scheduled Databricks notebooks before campaign performance degraded

Data Scientist - Markets and Markets — May 2022 - Feb 2023
Azure Databricks, Azure SQL, Azure Blob, Power BI, Python, SQL, Excel

- Uncovered a $20M adjacent market opportunity ($5M projected 2-year revenue) for a Fortune 500 client by building a lookalike segmentation model scoring 200+ candidate verticals on tech-stack and customer-base overlap
- Surfaced emerging technology trends ahead of traditional market reports by building an NLP pipeline extracting category signals from developer forums, tech blogs, and patent filings, giving clients first-mover advantage in GTM timing decisions
- Increased segment scoring cadence from bi-annual to quarterly across 10,000+ company profiles (k-means, cosine similarity) by building automated ETL pipelines, giving clients 2-4x more current expansion intelligence without ad-hoc re-engagement
- Informed go/no-go market entry decisions for 5+ hypotheses per client engagement by building bottom-up TAM models and adoption curves (Prophet, statsmodels) projecting 5-to-10-year revenue potential across adjacent ecosystems

Data Analyst - Zion Market Research — May 2019 - Apr 2022
Azure Databricks, Azure SQL, Power BI, R, Python, SQL, Excel

- Validated a 14-point lift in purchase intent across 3 campaigns by designing causal exposed-unexposed studies (n=12,000+) with matched controls, informing $2.5M in media budget reallocation
- Classified 80,000+ survey responses into 15 brand perception themes in under 2 hours using NLP (TF-IDF, LDA), replacing a 3-week manual coding process
- Delivered needs-based segmentation (LCA, k-means) across 10,000+ respondents into 5 clusters for a digital banking client, directly adopted into targeting rules that reduced their reported acquisition cost by 18%
- Isolated a winning product positioning outperforming control by 19% on purchase intent by designing monadic A/B tests across 6 messaging variants (n=8,000+) with Bayesian significance testing
- Cut reporting effort by 40 hours/month by building automated data pipelines for weekly brand health tracking across 5 concurrent accounts

Academic Projects

Customer Lifetime Value Prediction & Segmentation Engine *(Python, XGBoost, SHAP, Streamlit, Docker, MLflow, DVC, Optuna)*

- Engineered a hybrid probabilistic-ML pipeline (BG/NBD + Gamma-Gamma + Optuna-tuned XGBoost) on 1M+ e-commerce transactions, halving CLV prediction error over the probabilistic-only baseline (MAE £673 to £334, R^2 0.965)
- Built a CLV scoring dashboard with SHAP explanations and What-If simulator for campaign ROI estimation, backed by PSI drift monitoring, 94 pytest tests, and DVC-versioned artifacts **GitHub**

Online Experimentation Platform Simulator *(Python, SciPy, statsmodels, NumPy, Plotly, Streamlit)* **GitHub**

- Demonstrated that daily A/B test monitoring inflates false positives from 5% to 25-30% on 50K+ simulated sessions, and implemented mSPRT sequential testing enabling valid anytime checks
- Reduced user exposure to underperforming test variants by implementing Thompson Sampling bandit that dynamically routes 85%+ of traffic to the winning treatment in real time